

TABLE III
Sensor Types and Measurement Principles

Measuring principle	Sensor type	Common materials used
Chemoresistors	Metal oxides	Doped tin-dioxide, zinc oxide and tungsten trioxide
	Conductive polymers	Polypyrrole, polyaniline, polythiophene
	Conductive oligomers	Short-chain length conductive polymers
	Conductive-material-loaded polymers	Carbon particles in non-conductive polymers
	Self-organising films	Discotic liquid crystals
	Macrocyclic compounds	Phthalocyanines
Mass sensitive devices	Surface and bulk acoustic devices	Poly (siloxanes), charged polyelectrolytes, fluoropolymers, polyalkanes, hydrophilic polymers, calixarenes and other cavitands, chiral polymers, metal phosphonates, conducting polymers, lipids, self-assembled monolayers
Field-effect devices	Metal insulator semiconductors or metal-oxide semiconductors	Catalytic metals such as palladium, platinum and iridium, as well as other metals
Electrochemical devices	Oxidation or reduction catalysts	Precious metal and carbon with concentrated liquid electrolyte and counter electrode
Pellistors	Catalytic combustion devices	Platinum, palladium or their alloys
Fibre-optic systems	Polymer bids on fibre-optic bundles	Fluorescent and solvchromic dyes to detect solvation of vapours in the polymer
Intrinsic molecular parameters	Mass spectrometer, ion mobility or pulsed spectroscopy instrumentation	

Olfaction sensors have a wide range of transduction mechanisms. But at the output of these sensors, an electrical signal (say, voltage or current) proportional to the gas/liquid exposure on the sensor surface is required. Interface circuits convert the sensor output parameters into an electrical signal for further processing.

In chemo resistor sensors a change in resistance is obtained on exposure to the odour particles. A voltage divider or a Wheatstone bridge circuit can be used as the basic interface circuit. Interfacing circuits of quartz crystal microbalance or surface acoustic wave devices are required to measure shift in the resonance frequency.

The interfacing circuits are followed by the signal-conditioning block, which is basically analogue conditioning of the electrical signal through five sequential stages—buffering, amplification, filtering, conversion and compensation.

Data acquisition involves gathering signals from measurement sources, i.e., sensors, and digitising the signal for storage, analysis and presentation on a personal computer (PC). Data

acquisition systems are available in many different PC technologies for flexibility and application-specific customisation. Various options include peripheral component interconnect (PCI), PCI eXtensions for instrumentation (PXI), PCI Express, PXI Express, Personal Computer Memory Card International Association (PCMCIA), USB, IEEE 1394, parallel or serial ports for data acquisition in test, measurement and automation applications.

Signal pre-processing. Signal pre-processing means extraction of relevant information from the responses obtained and preparation of this information for multivariate pattern analysis. The major aspects of pre-processing are baseline identification and manipulation/determination, compression and normalisation or final preprocessing.

Baseline is the sensor response to a reference analyte and proper accommodation of the same into the actual sensor response is expected to provide compensation due to drift, scaling and enhancement of contrast and better identification. Three different techniques of baseline manipulation are differential, relative and fractional.

Compression vector or a fingerprint is extracted by reducing the number of descriptors. Normalisation is the final leg of pre-processing where techniques are applied to operate on the sensor signal to compensate for sample-to-sample variations due to change in analyte concentration and drift in the sensors. Alternatively, the entire database of a single sensor can be operated upon and scaling of sensors effects. The former technique is known as local method and

the latter as global method.

The responses generated by an array of olfaction sensors are processed using a variety of techniques. The pattern recognition engine may be developed using both parametric and non-parametric methods. Parametric methods are commonly referred to as statistical approaches and are based on the assumption that the sensor data set may be represented as a probability distribution function. The non-parametric methods do not assume any probability distribution function and deal with biologically inspired techniques, viz, artificial neural networks and expert systems.

A MOS-based electronic nose has been developed for tea aroma characterisation at C-DAC, Kolkata. The e-nose setup is shown in Fig.3.

The following e-noses are available in the market: FOX 2000/3000/4000, PEN 3, GDA 2, i-PEN, Gemini, Cybernose, ZNose 4200/4300/7100, Astree, Smart Nose 2000 and LibaNose 2.1.